Process for production of high density/high performance binderless boards from whole coconut husk: Part 1: Lignin as intrinsic thermosetting binder resin

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Abstract

Coconuts are abundantly growing in coastal areas of tropical countries. The coconut husk is available in large quantities as residue from coconut production in many areas, which is yielding the coarse coir fibre. The husk comprises ca. 30 wt.% coir fibres and 70 wt.% pith. Both fibre and pith are extremely high in lignin and phenolic content. The lignin is typically for monocotyledonous plants rich in syringyl with appreciable amounts of *p*-hydroxyphenyl units. The coir fibre is composed for at least one third of Klason lignin while lower molecular weight phenolics can be found as extractives in considerable amounts, especially in younger nuts. The thermal behaviour of the original (chemically unmodified) lignin in the plant tissues at temperatures above 140 °C, where it melts and shows thermosetting properties, has been investigated. This property of the coconut husk lignin was explored for application as intrinsic resin in board production, utilising whole fresh husks. Based on this concept, a simple and efficient technology has been developed to produce high strength–high density panels, without addition of chemical binders. Technical details will be reported in following papers.

Introduction

Coconut palms (*Cocos nucifera* L.) are abundantly growing in coastal areas of all tropical countries and its wide variety of products are being applied in food and non-food products. Its nutritious nuts represent a major form of livelihood for millions of people (Ohler, 1984, Ohler, 1999, Batugal et al., 1998). Primarily, the palms are grown for the oil-rich copra comprised inside the coconuts (Fig. 1). In a mature coconut, the white meat (28 wt.%) is surrounded by a hard protective shell (12 wt.%) and a thick husk (35 wt.%) (Thampan, 1991). This husk surrounding the large seed is constituted of 30 wt.% fibre and 70 wt.% pith material. The coarse coir fibres are traditionally extracted by various retting and decortication procedures from the husk for the production of ropes and yarns, mats, brushes and padding of mattresses (Kirby, 1963). Especially, Southern India and Sri Lanka have a long standing tradition and are the major production areas of a wide range of diversified coir products (Van Dam, 2002a).

World wide 40 to 50 million tons of coconuts are grown annually (FAO statistics, 2003), producing 15 to 20 million tons of husks. For coir fibre and fibre products as a commodity, a productivity is estimated of around 700,000 tons per year (FAO statistics, 2003), which is only a fraction of the otherwise wasted raw material. Since only a small part of the available husk material is being utilised new outlets are of interest. For the production of paper pulps the extreme high lignin content of coconut husk will require high amounts of chemicals and give lower cellulose yields than conventional wood pulps.

The aim of this research program (Common Fund for Commodities: Coir based building and Packaging materials, CFC/FAO-IGHF/11) is the development of an efficient process for production of board materials from the whole coconut husk without addition of chemical binders or additives (Van Dam, 1997, Van Dam et al., 2002b). A technology is being designed to transform the wasted residues into building boards, utilising the available high lignin content in both pith and coir fibre (Fig. 2).

The production of binderless board material is of interest from both ecological and economical point of view. The use of renewable resources as CO_2 neutral raw materials will play a key role in sustainable developments. Substitution of expensive synthetic resins by the intrinsic lignin binders present in lignocellulosic fibre raw materials will counteract the depletion of fossil resources. By omitting synthetic resins in the board production process, emissions of toxic components (e.g. formaldehyde) are prevented.

The use of lignin in wood adhesive formulations has been investigated in the past and described in numerous patents and papers, however, so far with limited commercial impact. Generally, the reactivity of the lignin released from lignocellulose after chemical and/or thermo-mechanical extraction processes is insufficient for complete substitution of synthetic resin. Steam explosion and organosolv lignins for that purpose would be more reactive, as compared to kraft lignin or lignosulfonates (Ono and Sudo, 1989, Lora et al., 1989). To prepare adhesive resins from these lignins, generally, methyolation, or preferentially phenolation is required. It has been demonstrated that up to a maximum of 50% replacement of phenolic resin (PF) by (soda) lignins is possible without affecting board properties (Roy et al., 1989).

A number of papers describe the production of binderless panels manufactured with steamed fibers of various origin (Wilcox, 1953, Nelson, 1973, Fadl et al., 1977, Meyers, 1978, Suchsland et al., 1987, Hsu, 1986, Suzuki et al., 1998, Angles et al., 1999). The thermo-mechanical pre-treatments of the fibres have large effect on the properties of the panels. A number of chemical changes are known to occur. Autohydrolysis of the constituent cell wall polymers in the lignocellulosic fibres by the steam explosion has been described. Especially hemicellulose polysaccharides (pentosans) degrade under these elevated temperature conditions to form furfural (Gosselink et al., 1995). Under certain conditions lignin is reported to melt but may show uncontrolled depolymerisation and rearrangements by condensation reactions (Lin and Dence, 1992, Suzuki et al., 1998, Angles et al., 1999). Another approach has been to enhance lignin reactivity by enzyme catalysed oxidation (laccase) (Felby et al., 1998).

The suitability of coconut fibre and pith for board production have been studied separately utilizing phenol-formaldehyde and urea-formaldehyde resins (Fuentes et al., 1997, Viswanathan and Gothandapani, 1999). Satisfactory boards could be produced on a laboratory scale and find application in India as wood substitute (Coir Board of India, 2003).

The extracted coir pith was studied for binderless board production by hot pressing (Mari, 1996) and verified the effects of chemical treatments and processing conditions. It was concluded that acid treatment would be required to meet the minimum technical specifications for panels.

Steam explosion of coconut husk as pre-treatment for fibre isolation was patented (Van Vreeswijk, 1963) and in a preliminarily study found suitable for binderless high density board production (TNO/Bouwcentrum, 1979). Based upon this concept, study of a technology for whole husk processing into building boards was undertaken.

For use as a binder, the melting behaviour of lignin and the subsequent cross-linking at elevated temperatures are the key issues to address. The lignin rich pith tissue is found to form a coherent and thermosetting matrix during board manufacturing, gluing the fibres together. To understand the thermal conversion of the husk material during board manufacturing, the raw material has been analysed in detail for its composition and properties. Since the maturity of the husk at harvest is expected to influence the processing, nuts of various ages were compared. Both fibre and pith have been investigated by means of thermal analysis, viz. TGA and DSC.

Coconut husk materials

Whole coconuts of the Rennel Island Tall (RIT) and Laguna Tall (LAGT) varieties were obtained exclusively from the Zamboanga Research Centre of the Philippine Coconut Authority. The freshly harvested nuts of different maturity (6, 7, and 11 months) were shipped by air directly and used as fresh as possible. The nuts were stored at 4 °C before further processing.

SEM photography

SEM micrographs were made from fracture surfaces of the coconut husks using a Jeol JSM-5600 LV scanning electron microscope.

Chemical composition of coir husk, fibre and pith

Samples were

Husk morphology

The coconut husk or seed coat tissue (mesocarp) consists of fibrous matter containing sclereids with numerous ramiform pits (Esau, 1977), which is surrounded by a thin waxy skin (epicarp) (Fig. 1). The coir fibres are embedded in porous and elastic cork-like parachymatous tissue called pith (Fig. 3A). Freshly opened husks are of light colour but darken soon upon drying in the air. The soluble sugars and tannin like organic components then become insoluble (Sengupta et al., 1949).

Only a small

Conclusions

Both fibre and pith fractions of coconut husk are of similar composition and relatively rich in lignin. The amount of extractable components in pith tissues is, however, substantially higher. The lignin in coconut husk is rich in syringyl and characteristic for a monocotyledonous plant. The behaviour of the husk material at elevated temperatures shows typically an exotherm starting at 140 °C, which disappears in a second heating. This irreversible softening is indicative for thermosetting

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Process for production of high density/high performance binderless boards from whole coconut husk: Part 2: Coconut husk morphology, composition and properties

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Abstract

For production of compression moulded boards from whole coconut husk the auto-adhesive properties are derived from the intrinsic high lignin content. Since the properties of manufactured boards for a large part will depend on the input husk material these properties are studied here. Husks of different varieties and maturity have been investigated for their contents of fibre and pith. Relevant physical properties and chemical composition were determined. For accurate determination of tensile strength and stiffness of coir fibre, its density (ρ_{coir} = 1.2–1.3 g/cm^3) and cross-sectional area were determined. Both thickness and longitudinal swelling of the fibre were measured to assess the effects of water absorption.

The constituent fibre and pith fractions of coconut husk have been chemically analysed for polysaccharide (cellulose, pectin and hemicellulose) and lignin contents. Detailed carbohydrate analysis of the fibre indicated a glucose (cellulose) content of approximately 33% in the cell wall, together with 12% xylose as the major component of the hemicellulose fraction. Coir pith only contains 16% glucose. The effects of maturing on the chemical composition of the husk is analysed and a significant increase in glucose (cellulose) is observed, while for other sugars no dramatic change in relative amounts can be measured.

It was concluded that only slight differences can be observed between mature coconut husks of different origin, and all are suitable as input feedstock for boards production. More effects can be expected from the age of the nuts.

Introduction

The suitability of coconut husks as raw material for binderless board production and the performance of the produced board in, e.g. building applications is in particular determined by mechanical and water absorbing characteristics. These characteristics are depending on the properties of the constituent tissues of coconut husk, i.e. coir fibre and pith. The relevant properties to be examined include morphology, chemical composition and the mechanical and physical properties.

Like other crops, coconuts show a wide diversity in size, weight, shape and colour, depending on genetic variety and maturity of the nut at harvest (Ohler, 1999). To assess the suitability of coir husks of different types and age as feedstock for conversion to compression moulded panels and board products different husk types have been evaluated. Especially, the constituent coir fibres are expected to maintain, more or less, their initial structure and properties during husk pre-treatment and board manufacturing. Therefore, mechanical and water absorption characteristics of coir fibres will be indicative for their contribution to the board properties.

The relevant mechanical properties of coir fibre in board, i.e. tensile strength and stiffness, are addressed in this study. Since coir fibres have oval cross-sections, the area cannot simply be determined by microscopic monitoring of the fibre before tensile testing. A method to determine the cross-sectional area of coir fibres is discussed. The water absorption capacity of coir fibres, and especially the resulting swelling, is of influence for the dimensional stability of the board during and after manufacturing. The thickness and longitudinal swelling of coir fibres have been investigated in this research. Important board properties are related to its density. Therefore, the density of the starting husk material and coir fibres is addressed as well.

In contrast to the coir fibres, the weak pith tissue will be torn apart during mechanical pre-treatment and severely condensed during board manufacturing. As a consequence, from the initial pith properties no board properties can be derived. Therefore, no mechanical or physical tests have been performed on the pith as it is available in the fresh coconut. In order to gain understanding of the expected chemical changes during board manufacturing thermal conversion processes, both fibre and pith have been chemically analysed for their composition.

The aim of this CFC-funded research program is the development of an efficient process for production of board materials from coconut husk without addition of chemical binders or additives (van Dam et al., 2004a, van Dam et al., 2004b).

Coconut husk and fibre materials

Whole coconuts of the varieties Tagnanan Tall (TAGT), Baybay Tall (BAYT), Laguna Tall (LAGT), Catigan green dwarf (CATD), Rennel Island Tall (RIT) and Agta Tall (AGAT) were obtained exclusively from the Zamboanga Research Centre of the Philippine Coconut Authority. The freshly harvested nuts of different maturity (6, 7, 11 and 13 months) were shipped by air directly and used as fresh as possible. The nuts were stored at 4 °C before further processing.

Fibres were manually extracted from the fresh

Coir husk anatomy

Very similar morphological structures are observed at micrometer scale by detailed electron microscopic examination of fibre and pith tissues from husks of different coconut cultivars, despite their nuts with different exterior appearance and contents of fibre and pith. The varieties studied include the mature husks (11 months old) of Rennel Island Tall (RIT), Tagnanan Tall (TAGT), Baybay Tall (BAYT), Laguna Tall (LAGT), Catigan green dwarf (CATD) and Agta Tall (AGAT). Significant differences

Conclusions

The influence of cultivar on morphological and structural details of mature coconut husk, fibre and pith, the mechanical properties of the coir fibres (water uptake, swelling and tensile strength) and the chemical composition are very small. The effect of fibre cell wall thickness on mechanical properties seems evident when comparing the values found for AGAT and RIT fibres. The differences between husk varieties does not seem to affect the suitability of husk materials as raw material for

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